DCU

DUBLIN CITY UNIVERSITY

SEMESTER ONE EXAMINATIONS 2012

MODULE: **(Title & Code)**	CA320 Computability & Complexity
COURSE:	BSc. in Computer Applications (SE Stream)
YEAR:	3
EXAMINERS: **(Including Telephone Nos.)**	Dr. J. Power, Dr. F. Bannister, Dr. D. Sinclair, Ext no. 5510.
TIME ALLOWED:	2 hours
INSTRUCTIONS:	Answer 4 Questions. At least 1 Question **must** be from Section A. All questions carry equal marks.

Please do not turn over this page until instructed to do so

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones

Section A

QUESTION 1 [TOTAL MARKS: 25]

1(a) [5 Marks]

In the Haskell Programming language what is the difference between *tuples* and *lists*?

1(b) [10 Marks]

Let a triangle be represented by a tuple where each entry in the tuple represents the length of a side. Write a Haskel function that determines if the tuple represents a right-angle triangle, an equilateral triangle (in which all sides have the same length), an isosceles triangle (in which only two sides have the same length) or a scalene triangle (in which all side have different length).

1(c) [10 Marks]

A set is an unordered collection of distinct elements. In Haskell we can represent a set as a list in in which an element is not duplicated and the order of the elements in the list is not important. Hence the set $\{3, 1, 8\}$ could be represented by the list [8,3,1]. Write Haskell functions that calculate the union and intersection of two sets that are represented as lists. The union of two sets represented as lists should be a list that contains all the elements in both lists, and the intersection of two sets represented as lists is a list that contains all the elements common to both lists. Ensure that elements are not duplicated in the resulting lists.

[End of Question 1]

QUESTION 2 [TOTAL MARKS: 25]

2(a) [5 Marks]

In the context of Haskell, what is meant by *lazy evaluation* and give an example of it.

2(b) [5 Marks]

Write a *data type* for a binary tree.

2(c) [15 Marks]

A reverse *monkey puzzle sort* of a list is performed by storing the elements of the list into a binary tree and then traversing the tree so that the nodes of a subtree are visited in the order right child, parent and then left child. Write a reverse monkey puzzle sort in Haskell.

[End of Question 2]

Section B

QUESTION 3 [TOTAL MARKS: 25]

3(a) [5 Marks]

In the study of Computability & Complexity, why do we focus on languages?

3(b) [5 Marks]

What is the significance of the Pumping Lemmas for Regular languages and Context-free languages?

3(c) [15 Marks]

Convert the nondeterministic finite automaton describes the the following transition function (in tabular form) into a deterministic finte automaton that recognises the same language.

q	$\delta(q,a)$	$\delta(q,b)$	$\delta(q,\Lambda)$	$\delta^*(q,a)$	$\delta^*(q,b)$
0	$\{0\}$	$\emptyset$	$\{1,3\}$	$\{0,1,2,3,4\}$	$\{3,4\}$
1	$\{2\}$	$\{4\}$	$\emptyset$	$\{2\}$	$\{4\}$
2	$\emptyset$	$\{1\}$	$\emptyset$	$\emptyset$	$\{1\}$
3	$\{4\}$	$\{3\}$	$\emptyset$	$\{4\}$	$\{3\}$
4	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

[End of Question 3]

QUESTION 4 [TOTAL MARKS: 25]

4(a) [5 Marks]

Formally define a *context-free grammar.*

4(b) [10 Marks]

Formally describe the simplest automaton that can recognise a context-free grammar.

4(c) [10 Marks]

Construct an automaton (of the type indicated in part (b)) that recognises the language $L = \{ww^r | w^r$ is the reverse of the string $w\}$ over the alphabet $\Sigma = \{a, b, c\}$.

[End of Question 4]

QUESTION 5 [TOTAL MARKS: 25]

5(a) [5 Marks]

Formally define a Turing machine.

5(b) [10 Marks]

Prove that language $HALT_{TM} = \{\langle M, w\rangle | M$ is a Turing machine and M halts on $w\}$ is undecidable. What are the general implications of this result?

5(c) [10 Marks]

Prove that any language accepted by a nondeterministic Turing machine can also be accepted by a deterministic Turing machine, but that the nondeterministic Turing machine accepts the language in less steps.

[End of Question 5]

[END OF EXAM]